

Which of the following statements accurately describes the stereochemical outcome of the Strecker synthesis of alanine?

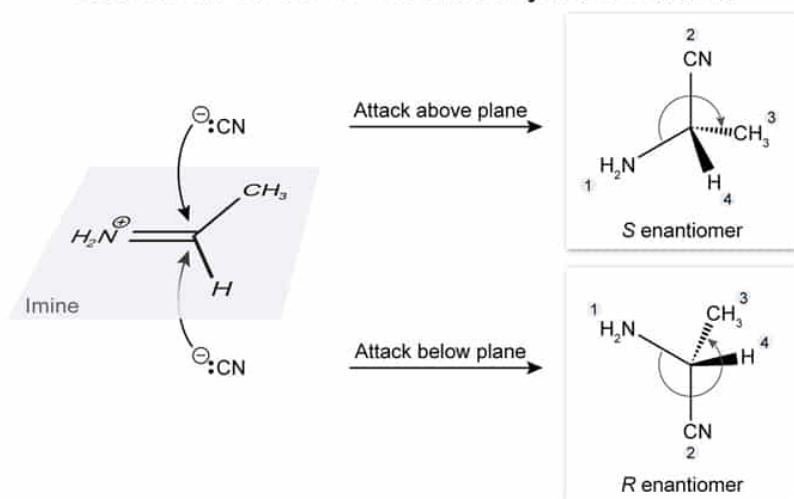
- ☐ A. The product is L-alanine. [30%]
- ☐ B. The product is D-alanine. [10%]
- ✔ ☒ C. The product is a mixture of L- and D-alanine. [56%]
- ☐ D. Alanine does not contain a stereocenter. [2%]

Correct

55% answered correctly

Explanation:

Stereochemical outcome of the Strecker synthesis of alanine



Because the imine intermediate is planar and has no stereochemistry, the product is a mixture of L- and D-amino acids.

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The **Strecker synthesis** is used to make α -amino acids from an aldehyde using ammonium chloride (NH_4Cl) and potassium cyanide (KCN). The first step of the reaction proceeds with protonation of the carbonyl oxygen by ammonium (NH_4^+), followed by nucleophilic attack of the carbonyl carbon by ammonia (NH_3), resulting in dehydration and imine formation. In the second step, a cyanide anion is

added to the imine to form an aminonitrile. Finally, in the third step, the nitrile ($R-CN$) nitrogen is protonated, and two water molecules add to the nitrile carbon in succession, eliminating ammonia from the nitrile and **forming the carboxylic acid** of the α -amino acid.

Alanine can be synthesized via the Strecker synthesis from acetaldehyde, NH_4Cl , and KCN . Acetaldehyde does not have a stereocenter, and the **imine intermediate** also does not have a stereocenter and is a **planar** molecule, allowing **nucleophilic attack** to occur from either above or below the plane. Therefore, the reaction is **not stereospecific** and the product will be a **mixture of L- and D-alanine**. The same outcome is obtained for amino acids produced by the **Gabriel synthesis** because the final step (**decarboxylation**) is not stereospecific.

(Choices A and B) Because the reaction proceeds through the formation of a planar intermediate, the product will be a mixture of L- and D-amino acids rather than just one or the other.

(Choice D) The α -carbon of alanine, as well as **all amino acids** except glycine, contains four different substituents, making it a **stereocenter**.

Educational objective:

The Strecker synthesis is used to generate α -amino acids from an aldehyde using ammonium chloride (NH_4Cl) and potassium cyanide (KCN). Because the imine intermediate formed during the reaction is planar (no stereocenters), nucleophilic addition can occur from either above or below the

plane. Therefore, the Strecker synthesis is not a stereospecific reaction and produces a mixture of L- and D-amino acids.

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Functional Groups and Their Reactions